VF-F03 Finish Adhesion

Test Method for Finish Adhesion	Test Method #: VF- F03 REF # : TM-R- 062
Date Issued: Dec 2, 2024 Revision # :3	Reason for Revision: Revise samples quantity from five to three

PURPOSE AND SCOPE: To measure the level of adhesion between a surface finish or coating and the substrate onto which it is applied. Test samples are prepared using controlled conditions and techniques and are evaluated using the 180° peel strength (T-peel) technique.

REFERENCES: DIN 53273 / 53578, ISO 11644

EQUIPMENT AND MATERIAL:

1. Coating device, for applying the primers and adhesives to specific uniform thicknesses. If available, a metered coating bar should be used, If a brush is to be used, care must be taken to ensure that a uniform, continuous application of adhesive is used.
2. Air-circulating oven, capable of maintaining a temperature of ± 1°C.
3. Thermocouple/surface pyrometer, capable of measuring the surface temperature of a part to ±1°C.
4. Press, capable of applying a pressure of 2-4kgf /cm^2 over the entire bond surface that is each cm^2 of the bond sample must have a load of 2-4 kgf /cm^2 applied to it. For example, a test sample which is 10 cm* 10cm, requires a load of 200-400 kgf.
5. Clean, lint-free cloth
6. Cleaning solvents, toluene free
7. Wire wheel buffing machine or sanding belt
8. Balance, capable of weighing to 0.01 grams.
9. Clean, dry stirring rod
10. Tensile testing machine, capable to capture graph.
11. Clamping jaws for tensile tester
12. Steel scale, capable of measuring to 0.5mm.
13. Full grain leather, polyurethane-coated leather, painted midsoles, and other coated materials.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. Test substrates should be die or hand cut 150 ± 50 mm in length and 25 ± 5 mm width.
3. Surfaces of test substrates should be cleaned and buffed according to the flow process set up by VF. If this is not available, prepare the surfaces according to the directions of the adhesive /primer supplier.)
4. Cleaning must always be performed with a clean, lint-free cloth and uncontaminated solvent, thorough drying of the surface is required prior to proceeding to the next step.
5. Priming should be performed according to the flow process set up by VF team, (If information concerning a specific material or combination of materials is not available, prime the surfaces according to the directions of the primer supplier). When bonding rubber, cover the entire bonding

surface with a thin, continuous application of the halogenation primer. Allow the primer to dry for 35 ± 5 minutes at 23 ± 2ºC, prior to applying the adhesive.

Note : ***When halogenation priming is performed ,the primer is not to be used until 20 minutes after the addition and initial mixing of the halogenation power .The pot life of the primer is 2 hours .Any primer which has not been used within 2 hours must be discarded.***

6. Two-component adhesives should be prepared by adding 4.0± 0.1% hardener by weight (unless otherwise specified), following addition of the hardener, the adhesive should be thoroughly mixed by hand stirring for at least 60 seconds, upon completion of mixing, the adhesive must be allowed to remain unused for 10 minutes, adhesive should then be applied uniformly and continuously to the entire bonding surface, preferably using a metered coating bar. Any adhesive which has not been used within 30 minutes of mixing must be discarded.

Note: ***Because the samples will need to be inserted into the jaws of a tensile tester, the samples should be assembled such that no bond is present at one end of the test samples for a distance of 35 ± 5mm, this can be most easily accomplished by not applying adhesive to this portion of the test sample.***

7. Dry the adhesive at 23 ±2ºC for 35 ± 5 minutes (or oven dry at the recommended conditions of the adhesive supplier), before heat activation the adhesive.
8. Place the bond samples in an oven at 70±1ºC until the adhesive surface reaches 60 ± 2 ºC. The surface temperature is measured using a surface pyrometer.
9. Assembled and press the bond samples immediately (<10 seconds) upon removal from the oven. Pressing must be undertaken for 10 seconds using a pressure of 2-4kgf/cm2. That is, each cm2 of the bond sample must have a load of 2-4 kgf applied to it. For example, a test sample which is 10cm*10cm requires a load of 200-400 kgf.
10. Prepared bond samples must be conditioned at 23±2 º C and 55 ±10% humidity for a minimum of 24 hours prior to proceeding to subsequent evaluations.
11. Following conditioning, die or hand cut test samples which are 15 cm in length by 2.5-3.0 cm in width.

REPETITIONS: Prepare three (3) bond samples per material combination, if aging/Hydrolysis evaluations are to be undertaken, three un-aged (control) samples and three aged samples must be prepared per adhesive or material combination

Note: If the test material is considered directional, three (3) samples should be prepared in both the length and width directions.

TEST PROCEDURE:

A: Dry Test (Un-aged)

1. Mark the specimen at the equidistant location - every 2 cm along the entire length of the sample.
2. Measure the width of the test sample to the nearest 0.05mm on every 2 cm mark. (**W**).
3. Zero and calibrate the tensile testing machine, set the crosshead speed to 100mm/minute, the jaw gap to approximately 3cm, and the load range to 10 kgf.
4. Centrally clamp approximately 20mm of the rubber portion (or stretchy material) of the bond sample into the stationary jaw of the tensile tester. Centrally clamp approximately 20mm of the free end of the second substrate of the bond sample into the moving jaw of the tensile tester. The sample must be inserted into the jaws such that the bond line can be observed by the operator as peeling progresses.

Note: ***To help ensure that failure occurs between the surface coating or finish and the substrate onto which it applied, it may be useful to carefully cut through the surface finish or coating prior to initiating peeling.***

5. Operate the tensile testing machine at a speed of 100 mm/minute.
6. If peeling stops before two thirds of the total length, remove the force and carefully peel the test specimen by hand for about a millimeter and then apply the same force again.
7. Continue testing until the bond sample has been completely separated.
8. Remove the test specimen from the holder. Examine and record the type of bond separation corresponding to the maximum peeling load as:

Valid types of separation:

A – finish coats separating
B – removal of finish from substrate
C – surface of substrate tearing off

Invalid types of separation:

D – separation of the adhesive from the strip

E – separation of the adhesive from the finish

F – tearing down the middle of the finish

G – deep tearing in the substrate

9. Record the average peeling force, **kgf** for all valid test specimens in every 2 cm distance: **(F)**

B: Wet Test (Un-aged)

1. Submerge test samples in distilled water at 23.0 ±1.0°C for 2 hours.
2. Remove samples from water, and test immediately following the directions outlined in “A”) DRY test (Un-aged) above.

CALCULATONS:

Peel Strength (kgf /cm) = F / W

Where:

F: Breaking Force in kgf

W: Width of specimen in cm

REPORT:

1. Average peel strengths: kgf /cm
2. Types of separation